SWYNEX Technologies

Data Analyst Internship — Task 1 Submission Documentation

INTERN NAME	INTERN ID	TASK NAME	STATUS
Maria Shaikh	SWX-2026-002703	Data Cleaning	Ready to Submit

Task Objective: Select a raw dataset, identify missing values, duplicate records, incorrect data types, and formatting inconsistencies, then perform data cleaning using Python (Pandas) and document the output.

1. RAW DATASET OVERVIEW (`RAW_DATA.CSV`)

The table below illustrates the raw, uncleaned dataset containing errors like duplicated rows, whitespace errors, inconsistent capitalization, missing fields, and unformatted dates.

ID	Name	Age	Salary	Department	Join_Date
1	ali khan	25	50000.0	it	2023-01-15
2	Sara Ahmed	30	60000.0	HR	2022-05-20
1	ali khan	25	50000.0	it	2023-01-15 (Duplicate)
3	Ahmed Raza	NaN (Missing)	45000.0	finance	2023-03-10
4	Fatima	28	NaN (Missing)	IT	2021-11-12

2. CLEANED OUTPUT OVERVIEW (`CLEANED_DATA.CSV`)

Result after processing with the Python cleaning script:

ID	Name	Age	Salary	Department	Join_Date
1	Ali Khan	25.0	50000.00	IT	2023-01-15
2	Sara Ahmed	30.0	60000.00	HR	2022-05-20
3	Ahmed Raza	26.5 (Median)	45000.00	FINANCE	2023-03-10
4	Fatima	28.0	51666.67 (Mean)	IT	2021-11-12

3. DATA CLEANING PYTHON SCRIPT (`DATA_CLEANING.PY`)

```
import pandas as pd

# 1. Load Raw Dataset
df = pd.read_csv('raw_data.csv')

# 2. Remove Duplicate Records
df = df.drop_duplicates()

# 3. Impute Missing Values
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())
df['Department'] = df['Department'].fillna('UNKNOWN')

# 4. Standardize Text Formatting
df['Name'] = df['Name'].str.strip().str.title()
df['Department'] = df['Department'].str.strip().str.upper()

# 5. Convert Data Types
df['Join_Date'] = pd.to_datetime(df['Join_Date'])

# 6. Export Cleaned Dataset
df.to_csv('cleaned_data.csv', index=False)
print("Data Cleaning Completed Successfully!")
```

4. SUMMARY OF DATA CLEANING OPERATIONS

1. Duplicate Removal

Identified and dropped identical row entries (ID #1 duplicate) to prevent inflated statistics.

2. Missing Value Imputation

Replaced missing age with median (26.5) and missing salary with column mean (51,666.67).

3. Text Whitespace & Casing

Trimmed unnecessary spaces around names and departments; formatted names to Title Case and departments to UPPERCASE.

4. Datetime Formatting

Converted text date fields into standardized Pandas ISO Datetime values (YYYY-MM-DD).

5. GITHUB & LINKEDIN SUBMISSION LINKS

GitHub Repository Link:

`https://github.com/[your-username]/SWYNEX-Data-Cleaning-Preparation`

LinkedIn Post Link:

`https://www.linkedin.com/posts/[your-linkedin-username]/swynex-task1-submission`